Technology Stack

Date	7 July 2026
Team ID	
Project name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum marks	2 marks

Technology Stack

The Smart Agricultural Production Optimization Engine (OptiCrop) is developed using a modern technology stack to ensure reliability, efficiency, and scalability. The frontend is built with HTML, CSS, and JavaScript to provide a responsive user interface, while the backend uses Python and Flask to process user requests and integrate the machine learning model. MySQL is used for data storage, and GitHub is used for version control. Machine learning libraries such as Scikit-learn, Pandas, and NumPy are used for crop prediction and agricultural data analysis.

S.No	**Architecture Component / Layer**	**Technology Chosen**	**Justification / Purpose**
1.	Frontend / Client-Side	HTML5, CSS3, JavaScript	Create a responsive and user-friendly interface
2.	Backend / Server-Side	Python, Flask	Handle business logic and machine learning integration
3.	Database / Data Storage	MySQL	Store user information and agricultural datasets
4.	Cloud / Hosting / Deployment	Localhost / Flask Serve	Easy development and testing
5.	Version Control & CI/CD	Git & GitHub	Source code management and collaboration

6.	Third-Party APIs / Other Tools	Weather API, Scikit-learn, Pandas, NumPy	Weather information, machine learning, and data processing